Empirical Predictions of The Phantom Metric: Falsifying Dark Matter via High-Redshift Kinematics and Macroscopic Lensing

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Abstract

We present three falsifiable predictions of The Phantom Metric, a parameter-free saturated metric framework. The model derives galactic rotation curves and gravitational lensing solely from baryonic mass and a fundamental acceleration scale $a_0 = \frac{cH_0}{2\pi}$, rendering particulate dark matter mathematically redundant.

Prediction I: JWST observations of dynamically settled disk galaxies at $z > 3$ will exhibit strict rotational flattening, deterministically obeying $V_{calculated} = \sqrt{V_{baryonic}^2 \nu}$ with a universally locked stellar mass-to-light ratio ($\Upsilon_* \approx 0.46$). This proves that the geometric topological confinement effect is invariant to the galaxy's evolutionary epoch, requiring no retroactive dark halo assembly.

Prediction II: Euclid observations of morphologically relaxed galaxy clusters will yield Einstein ring radii (R_E) strictly dictated by the Macroscopic Saturation

Boundary, precisely defined as $R_E = R_{BB} = \sqrt{\frac{GM_{baryon}}{a_{0_{eff}}}}$ using only the observable $\sim 13\%$ baryonic mass fraction. For a specific benchmark cluster mass of $M_{bary} = 0.98 \times 10^{11} M_{\odot}$, the metric definitively predicts a lensing radius of ~ 11.00 kpc, demonstrating a 1:1 structural parity ($R^2 > 0.95$) with zero free parameters.

Prediction III: Next-generation surveys (e.g., Euclid, Roman) discovering uncharted isolated strong gravitational lenses will demonstrate an extreme $> 99\%$ structural parity ($R^2 > 0.99$) between the observed Einstein ring and the baryonic Macroscopic Saturation Boundary, perfectly matching the local SLACS observations with zero dark matter.

Confirmation of these numerical thresholds would constitute a definitive falsification of the standard ΛCDM dark matter paradigm.

1. Introduction

The standard ΛCDM paradigm interprets flat galactic rotation curves and excessive gravitational lensing as evidence for invisible particulate matter. However, recent formulations of *The Phantom Metric* [1] demonstrate that these anomalies emerge deterministically from topological saturation threshold governed by a strict topological scaling ratio, eliminating the need for free parameters. To empirically isolate this geometric framework from standard dark matter models, we present two definitive predictive bounds.

2. Prediction I: Baryon-Only Flattening in High-Redshift Galaxies (JWST)

Current Standard Model Paradigm: Standard cosmology predicts that early galaxies ($z > 3$) should exhibit fundamentally different rotational kinematics because their extended dark matter halos have not yet fully assembled. Consequently, classical Keplerian decline is expected at their outer radii.

The Phantom Metric Prediction: The topological saturation threshold function (v) is a fundamental geometric property of the spatial saturated metric framework, not a consequence of accumulating particulate mass. Therefore, geometric topological confinement effect is invariant to the galaxy's cosmic age.

Empirical Falsification Test: High-resolution rotation curves of dynamically settled disk galaxies at $z > 3$ observed by the James Webb Space Telescope (JWST) will exhibit strict rotational flattening identical to local late-type galaxies. The rotational velocity will deterministically obey:

$$V_{calculated} = \sqrt{V_{baryonic}^2 v}$$

This correlation will hold using a universally locked stellar mass-to-light ratio ($\Upsilon_* \approx 0.46$), proving that dark matter halo fitting remains mathematically obsolete regardless of the galaxy's evolutionary epoch.

Table 1: Kinematic Falsification at High Redshift ($z > 3$)

Parameter	Standard Model (ΛCDM)	The Phantom Metric
Outer Rotation Curve	Classical Keplerian decline expected	Strict rotational flattening identical to local galaxies
Rotational Velocity	Requires retroactive dark matter parameter fitting	$V_{calculated} = \sqrt{V_{baryonic}^2 v}$
Mass-to-Light Ratio	Treated as a free, adjustable parameter	Universally locked at $\Upsilon_* \approx 0.46$
Target Data Source	JWST (Unassembled Halos)	JWST (Invariant topological confinement effect)
Numerical Benchmark Target	Velocity remains unpredictable until a hypothetical "Dark Matter Halo" is manually fitted to the curve.	For a given $V_{baryonic}$, the outer velocity is deterministically locked by the topological saturation threshold (v) with 0 free parameters.

3. Prediction II: Macroscopic Lensing Boundaries in Uncharted Clusters (Euclid)

Current Standard Model Paradigm: The Euclid space mission is currently mapping massive, uncharted galaxy clusters, anticipating the need to map complex dark matter density profiles to account for strong gravitational lensing radii.

The Phantom Metric Prediction: For any newly discovered, morphologically relaxed galaxy cluster ($\eta \approx 1$), macroscopic photon deflection paths will not follow continuous 3D mass profiles. Instead, light bending is deterministically routed along the Macroscopic Saturation Boundary (R_{BB}).

Empirical Falsification Test: Euclid's strong lensing data for relaxed clusters will yield an Einstein ring radius exactly equal to:

$$R_{BB} = \sqrt{\frac{GM_{baryon}}{a_{0eff}}}$$

This bound will be achieved strictly by utilizing the observable ~ 13% baryonic mass fraction (X-ray gas and stellar mass) with 0% dark matter, demonstrating a strict 1:1 structural parity ($R^2 > 0.95$).

Table 2: Macroscopic Lensing Boundaries in Uncharted Clusters

Parameter	**Standard Model (ΛCDM)**	**The Phantom Metric**
Photon Deflection Path	Follows continuous 3D complex dark matter density profiles	Deterministically routed along the Macroscopic Saturation Boundary
Einstein Ring Radius	Calculated via non-baryonic halos	$R_{BB} = \sqrt{\frac{GM_{baryon}}{a_{0eff}}}$
Mass Input Required	~100% mass budget (requires invisible dark matter)	strictly by utilizing the observable ~ 13% baryonic mass fraction
Structural Parity	Requires arbitrary density parameters per cluster	Strict 1:1 structural parity ($R^2 > 0.95$) utilizing 0% dark matter
Numerical Benchmark (e.g., $\boldsymbol{M_{baryon} = 0.98 \times 10^{11} M_{\odot}}$)	Predicted radius varies wildly depending on the arbitrary Dark Matter fraction (f_{DM}) assigned.	Predicts an Einstein radius of exactly ~ 11.00 kpc, locking a 1:1 structural parity with zero free parameters.

4. Prediction III: Strong Lensing in Isolated Galaxies (Euclid / Roman)

Current Standard Model Paradigm: Standard cosmology assumes that massive elliptical galaxies acting as strong gravitational lenses are embedded in massive, invisible dark matter halos. Calculating the exact deflection paths of background photons requires complex, localized curve-fitting of these hypothetical dark matter profiles for each newly discovered lens.

The Phantom Metric Prediction: The Einstein ring radius is not a product of continuous mass curvature, but a direct physical manifestation of the Macroscopic Saturation Boundary (R_{BB}). For isolated galaxies ($\eta \approx 0$), the Macroscopic Saturation Boundary is determined strictly by the visible baryonic mass interacting with the baseline topological acceleration limit (a_0).

Empirical Falsification Test: Upcoming large-scale surveys (e.g., the Euclid mission and the Nancy Grace Roman Space Telescope) are projected to discover tens of thousands of new strong gravitational lenses. For any newly discovered isolated lens, the observed Einstein ring radius (R_E) will deterministically match the theoretical Macroscopic Saturation Boundary:

$$R_{BB} = \sqrt{\frac{GM_{baryon}}{a_0}}$$

This purely baryonic calculation will consistently yield a structural parity exceeding 99% ($R^2 > 0.99$), rendering the retroactive fitting of dark matter halos statistically and mathematically obsolete.

Table 3: Isolated Strong Lensing Predictions for Next-Generation Surveys

Parameter	**Standard Model (ΛCDM)**	**The Phantom Metric**
Photon Deflection Source	Dense, non-baryonic dark matter halos	Strict topological spatial saturation boundary
Prediction Methodology	Retroactive mass-profile fitting per galaxy	$R_{BB} = \sqrt{\frac{GM_{baryon}}{a_0}}$
Free Parameters	Multiple (Halo mass, concentration, density)	Zero
Statistical Falsification Threshold	Assumes high intrinsic scatter without dark matter	Predicts an extreme global correlation of $R^2 > 0.99$ based *only* on the visible baryonic payload.

References

[1] Haimovich, T. (2026). " The Phantom Metric: Resolving Dark Matter Anomalies in Galactic Kinematics, Strong Lensing, and Galaxy Clusters via Topological Constraints and Topological Metric Deformation (Parameter-Free)" Zenodo Digital Repository. 10.5281/zenodo.22661163.